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Pet Waste Tray

Abstract

A pet waste tray including a downward angled ramp of a unified one-piece construction with the tray that holds the front of the tray higher than the back of the tray to prevent a first piece of collected pet waste from falling off the tray when a subsequent piece of pet waste is collected. Additionally, the tray may include one or more holes in a back wall of the tray to allow a user to clean the tray from behind with a stream of water (e.g., from a garden hose) without splashing water and/or pet waste on the user.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates to an apparatus to collect pet waste. More particularly, the present invention relates to an improved pet waste collection tray that prevents collected pet waste from falling off the tray when additional pet waste is collected.

BACKGROUND OF THE INVENTION

[0002] Many pet owners use a combination of a tray and rake (or spade) to pick up their pet's solid waste (i.e., dog feces). The tray may be flat and surrounded by three walls. Namely, a back wall and two sidewalls that leave the front open to receive pet waste. Most commonly, a handle is attached to the rear of the back wall when the tray is fixed (e.g., the tray does not swing or move independently of the handle). The tray's front edge is usually flush with the most forward point of each sidewall.

[0003] The tray may include a vertical handle connected to the rear of the back wall to allow the owner to collect waste without bending over. The rake is usually used for uneven surfaces (e.g., grassy lawn) and the spade for flat surfaces (e.g. a kennel floor) to scrape waste onto the tray. The owner then dumps the waste into a bag, or other receptacle.

[0004] Many pet owners collect their dog's waste at the end of the day from a back yard (e.g., after work). By this time, a dog may have deposited feces at multiple locations in the yard.

[0005] A problem with this type of prior art tray is that when a first piece of solid waste has been collected and the tray is tilted forward to collect a second piece of solid waste, the first piece of waste often falls off the tray onto the ground. This is especially true when the waste is dry and cylindrically shaped, or when the surface that the waste is gathered from is uneven and requires the tray to be tilted almost vertically to get under the waste to prevent the waste from being smeared on the lawn.

[0006] Another problem with prior art pet waste collection trays is that when one rinses them with water from a garden hose, water mixed with pet waste often splashes off an interior tray wall and onto the user. Think of washing a spoon in a sink and having the water splash on you, but with much more water mixed with pet waste. Therefore, there remains a need for a pet waste collection tray that can be used to collect a second piece of pet waste without having an already collected first piece of pet waste falling off the tray, and that can be rinsed off without splashing a user with pet waste.

SUMMARY OF CERTAIN INVENTIVE ASPECTS

[0007] To overcome the aforesaid deficiencies in prior art pet waste collection trays, disclosed is a pet waste collection tray apparatus that includes a ramp that extends from the front of the tray in a downward angle, relative to the top surface of the tray, so that a first piece of collected pet waste will not fall off the tray's surface when a second piece of pet waste is collected onto the tray. The ramp is preferably made of a unified one-piece construction with the tray, and be made of metal (e.g., aluminum), plastic, or other suitable material. The tray's back wall may include one or more holes to receive a stream of water from behind to rinse off the tray's surface without splashing feces on a user. A substantially vertical handle may be attached to the rear of the back tray wall when the tray is fixed (i.e., the tray moves when the handle is moved). However, the handle may be attached at a different angle, or at a different location on the tray.

[0008] In a preferred embodiment, the invention comprises a tray for collecting pet waste, the tray further comprising a top surface, a back, a front edge, and a downward angled ramp; wherein the ramp is of a one-piece unified construction with the tray and extends downward from the tray's front edge to hold the front of the tray higher than the back of the tray to prevent a first piece of collected pet waste from falling off the tray's top surface when a second piece of pet waste is collected onto the tray. Additionally, the tray's top surface may include a raised portion (e.g., a ridge, a bump(s), etc.) to prevent a first piece of pet waste from falling off the tray by blocking it when the tray is tilted forward to collect a second piece of pet waste. For example, the raised

portion may include a “speed bump” like ridge that is substantially parallel to the tray's front. The ridge may be a singular solid length, a collection of lengths, or include staggered segments, a bump, or a pattern of raised bumps. The raised portion may be of a unified one-piece construction with the tray or be attached to the tray's top surface. Alternatively, or in combination with a raised portion, the surface may include one or more small cavities, indentations, pits, etc. to keep a first piece of collected waste from falling off the tray.

[0009] In an alternate embodiment, the invention comprises a ramp apparatus for a pet waste tray, wherein the ramp holds the tray's front higher than the tray's back end when the tray is on a flat surface to prevent a first piece of collected pet waste from falling off the tray when a subsequent piece of pet waste is collected. In this embodiment the ramp may not be of a one-piece construction with the tray, but instead is removably attachable to a pet waste tray's front edge by means of an engaging structure/mechanism (e.g. clamping, friction fit, screws, adhesive, etc.). Alternatively, the ramp may be of a unified one-piece construction with the tray. Further, the ramp may be made of the same or different material than the tray. For example, many fixed trays are metal, but the attachable ramp may be made of plastic or other suitable material.

[0010] In another alternative embodiment, the invention comprises a raised ridge for a substantially flat surfaced pet waste tray. The raised ridge is for the pet waste tray's top surface and is configured to prevent a first piece of pet waste, that has been collected onto the tray's top surface between the raised ridge and a back wall of the tray, from falling off the tray's top surface when a subsequent piece of pet waste is collected onto the tray's top surface. The ridge may be substantially parallel to the front edge of the tray to prevent collected pet waste from falling off the tray's surface when collecting a second piece of waste. The ridge may be located proximate to the front edge of tray and/or perpendicular to a side wall of the tray. In yet another alternate embodiment, the tray's surface may not be flat, but instead, in whole or in part, be folded/shaped to have ripples on it (e.g., like a RUFFLES® Original Potato Chips) to stop pet waste from falling off the tray when a second piece of waste is collected.

[0011] In yet another alternative embodiment, the invention comprises a back wall for a pet waste collection tray, wherein the back wall comprises a hole that is configured to receive a stream of water from behind the back wall (e.g., from a garden hose) to clean a surface of the pet waste collection tray and prevent water mixed with pet waste from splashing a user when the user cleans the tray. The hole(s) may be of a size (e.g., a diameter) large enough to accommodate a stream of water from a garden hose to allow the stream to substantially pass through the wall without the stream striking the back wall and/or the edges of the hole.

[0012] Other objectives and aspects of the invention will become apparent from the following detailed description, taken in conjunction with the accompanying drawings, which illustrate, by way of example, the features in accordance with embodiments of the invention.

[0013] To the accomplishment of the above and related objects, the invention may be embodied in the form illustrated in the accompanying drawings, attention being called to the fact, that the drawings are illustrative only, and that changes may be made in the specific construction illustrated, and the manner described, that are within the scope of the appended claims. Thus, the breadth and scope of the present invention should not be limited by any of the above-described exemplary embodiments.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0014] FIG. 1 is a top perspective view of a new pet waste tray and ramp.

[0015] FIG. 2 is a front view of the new pet waste tray and ramp in FIG. 1.

[0016] FIG. 3 is a back view of FIG. 2.

[0017] FIG. 4 is a right-side view of FIG. 2.

[0018] FIG. 5 is a left-side view of FIG. 2.

[0019] FIG. 6 is a top view of FIG. 2.

[0020] FIG. 7 is a bottom view of FIG. 2.

[0021] FIG. 8 is a perspective view of an alternate embodiment of an attachable ramp for a pet waste tray.

[0022] FIG. 9 is a top view of the ramp in FIG. 8.

[0023] FIG. 10 is a left side view of the ramp in FIG. 8.

DETAILED DESCRIPTION

[0024] The present invention solves the aforesaid problems by providing a new ramped tray for collecting pet waste that prevents a first piece of collected pet waste from accidentally falling off the tray onto the ground when a user collects a second piece of pet waste. These and other features, aspects and embodiments of the invention will be described in more detail below.

[0025] The presence of broadening words and phrases such as “one or more,” “at least,” “but not limited to”, or other like phrases, shall not be read to mean that the narrower case is intended, or required, where such broadening phrases may be absent.

[0026] The following detailed description is directed to certain specific embodiments of the invention. However, the invention can be embodied in a multitude of diverse ways as defined and covered by the claims. In this description, reference is made to the drawings wherein like parts are designated with like numerals throughout.

[0027] The objects and features of the present invention will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings. Understanding that these drawings depict only typical embodiments of the invention and are not to be considered limiting of its scope, the invention will be described and explained with additional specificity and detail using the accompanying drawings.

[0028] FIG. 1 shows a perspective view of a preferred embodiment of the ramped tray apparatus 10. The tray 10 includes a ramp 12 that extends in a downward angle from the front 14 of the tray's 10 top surface 16. The ramp 12 preferably is of a one-piece unified construction with the tray 10. The tray 10 preferably includes two sidewalls 18 & 20, and a back wall 22. The back wall 22 may include at least one hole 24 to allow water from a garden hose (not shown) to pass through the back wall 22 and onto the top surface 16 of the tray 10 to rinse off any stuck-on pet waste (not shown) without splashing wastewater on the user. The hole 24 is preferably large enough to permit a stream of water from a garden hose to pass through the hole 24 without splashing a user but may be a different size/configuration (e.g., many small holes).

[0029] Additionally, or instead of a ramp 12, the tray 10 may comprise at least one ridge 26, or other upward protruding feature (e.g. one or more bumps 28) on the top surface 16 of the tray 10 to prevent a first piece of collected pet waste from falling off the surface 16 when a second piece of pet waste is collected. The ridge 26 may be created by affixing an object to the tray's 10 top surface 16 (as shown), by folding the tray 10 itself to create a ridge 26 on the top surface 16, other one-piece construction with the tray 10. The tray 10 preferably includes a handle 30 that is attached to the rear of the back wall 22.

[0030] FIG. 2 shows a front view of the new pet waste tray 10. The tray 10 includes a downward angled ramp 12 that extends from the front 14 of the tray's 10 top surface 16. In a preferred embodiment, the tray 10 and ramp 12 are of a unified one-piece construction. The surface 16 is preferably surrounded by two sidewalls 18 & 20 and a back wall 22. The back wall 22 may include at least one hole 24 to allow a user to direct a stream of water from a garden hose through the hole 24 and onto the top surface 16 to clean it. A user may stand behind the tray 10 to rinse off the top surface 16, and since there is no wall opposite the back wall 20 a user can rinse the tray 10 without splashing dog feces onto him/herself. Alternatively, or in addition to a hole 24 in the back wall 22, a sidewall 18 & 20, or both, may include a hole (not shown) to rinse the tray 10.

[0031] The tray **10** may include a ridge **26**, or other raised portion(s), such as a bump **28**, to prevent a collected piece of pet waste from accidentally falling off the tray's **10** top surface **16** when a second piece of pet waste is collected.

[0032] FIG. **3** shows a back view of the tray **10**, including a handle **30**, back wall **22** and at least one hole **24**. The handle **30** may be attached to the rear of the back wall **22**, or other part of the tray **10**.

[0033] FIG. **4** shows a right-side view of the tray **10**, comprising a ramp **12** that extends in a downward angle from the front **14** top surface **16** of the tray **10**. The ramp **12** holds the front **14** of the tray **10** higher than the back **32** of the tray **10** (when resting on a flat surface), so that when pet waste is collected onto the top surface **16** of the tray **10**, it will not fall off when a second piece of pet waste is collected. Additionally, the tray's top surface **16** may include a raised ridge **26**, or other raised portion (e.g., a bump **28**), to keep collected pet waste from accidentally falling off the tray **10** when additional pet waste is collected. The raised ridge **26** may be of a one-piece unified construction with the tray **10**, or a separate apparatus attached to the tray's **10** top surface **16** (e.g., bolted or glued on). A handle **30** is preferably attached to the rear of the back wall **22** to allow a user to collect waste without bending over and/or manipulate the tray's **10** position to rinse off the tray **10** from behind, through a hole **24** in the back wall **22**.

[0034] FIG. **5** shows a left-side view of the tray **10**, comprising a ramp **12** that extends in a downward angle from the front **14** top surface **16**. The ramp **12** holds the front **14** of the tray **10** higher than the back **32** of the tray **10**, so that when pet waste is collected onto the top surface **16** of the tray **10** it will not fall off when a second piece of pet waste is collected. Additionally, the tray's top surface **16** may include a ridge **26**, or other raised portion (e.g., a bump **28**), to keep collected pet waste from accidentally falling off the tray **10** when additional pet waste is collected. A handle **30** is preferably attached to the rear of the back wall **22** to allow a user collect waste and/or manipulate the tray's **10** position to rinse the tray **10** from behind, through a hole **24** in the back wall **22**.

[0035] FIG. **6** shows a top view of the tray **10**, comprising a handle **30** attached to the rear of the back wall **22**. The back wall **22** may include a hole **24** useable to gush garden hose water (for example) through to clean the top surface **16** of the tray **10**. The tray **10** further comprises a ramp **12** that is preferably integral to the tray's material and of a unified one-pieced construction with the tray **10** that extends in a downward angle from the tray's **10** front **14** to hold the front **14** of the tray **10** higher than the back **32** of the tray **10** when the tray **10** rests on a flat surface. The top surface **16** may include a ridge **26**, or other raised portion (e.g., a bump **28**, or collection of them), and be surrounded by two sidewalls **18** & **20** and a back wall **22**.

[0036] FIG. **7** shows a bottom view of the tray **10**. The tray **10** comprises a ramp **12** that extends downward from the tray's **10** front that holds the front **14** of the tray **10** higher than the back **32** of the tray **10** to prevent a first piece of collected pet waste from falling off the tray **10** when a subsequent piece of waste is collected. The tray **10** includes two sidewalls **18** & **20** and a handle **30** that may be attached to the rear of the back wall **22**. The back wall **22** may include a hole **24** that is used to direct a stream of water through (e.g., from a garden hose) to clean the tray **10**. For example, a user can hold the handle **30** to turn the front **14** of the tray **10** away from the user and direct a stream of water through the hole **24** in the back wall **22** to rinse off the tray **10**. The tray's **10** top surface (not shown) may additionally include a ridge **26**, or bump **28**, to prevent a first piece of collected pet waste from falling off the tray **10** when a subsequent piece of pet waste is being collected.

[0037] FIG. **8** shows an alternate embodiment of a removably attachable downward angled ramp **34** for a pet waste tray (not shown). The ramp **34** comprises an engaging structure **36** at the top end of the attachable ramp **34** that is preferably integral to the ramp's **34** material that holds the ramp **34** to a tray's front end via friction created by opposing pressure of a top **38** and bottom **40** lip. The ramp **34** may be easily attached to a tray's front edge by pushing a tray's front edge between the lips

38 & 40 and into the angled ramp's **34** groove **42**. The ramp **34** will hold a flat tray's front higher than the tray's back to prevent a first piece of collected waste from falling off the tray when a subsequent piece of pet waste is collected.

[0038] FIG. **9** shows a front view of the removably attachable downward angled ramp **34**. The top of the angled ramp **34** comprises an engaging structure **36** that attaches to the front edge of a pet waste collection tray by way of friction created by the opposing pressure of a top **38** and bottom **40** lip.

[0039] FIG. **10** shows a left-side view of a removably attachable downward angled ramp **34**, that comprises an engaging structure **36** at the top of the ramp **34** that attaches to a pet waste collection tray's front edge by pushing the tray's front edge into the ramp's **34** slot **42** and is held in place by way of friction created by opposing pressure of a top **38** and bottom **40** lip.

[0040] Therefore, it must be understood that the illustrated embodiments have been set forth only for the purposes of examples and that they should not be taken as limiting the invention as defined by the following claims. For example, notwithstanding the fact that the elements of a claim are set forth below in certain combination, it must be expressly understood that the invention includes other combinations of fewer, more, or different ones of the disclosed elements.

[0041] Insubstantial changes from the claimed subject matter as viewed by a person with ordinary skill in the art, now known or later devised, are expressly contemplated as being equivalently within the scope of the claims. Therefore, obvious substitutions now or later known to one with ordinary skill in the art are defined to be within the scope of the defined elements.

[0042] The claims are thus to be understood to include what is specifically illustrated and described above, what is conceptually equivalent, what can be obviously substituted, and what incorporates the essential idea of the invention.

Claims

1. An apparatus, comprising: a tray for collecting pet waste, the tray further comprising a top surface, a back, a front, and a downward angled ramp; wherein the ramp is of a one-piece unified construction with the tray and extends downward from the tray's front to hold the front of the tray higher than the back of the tray to prevent a first piece of collected pet waste from falling off the tray's top surface when a subsequent piece of pet waste is collected onto the tray.
2. The tray of claim 1, further comprising three walls around the top surface, including a back wall and two sidewalls.
3. The tray of claim 1, further being fixed and not independently moveable from a handle attached to a back wall of the tray.
4. The tray front of claim 1 further being higher than the tray back when the tray is placed on a flat surface.
5. The top surface of claim 1 further comprising a raised ridge to stop a first piece of collected pet waste from falling off the top surface of the tray when a second piece of pet waste is collected.
6. The raised ridge of claim 5 further comprising a length.
7. The length of claim 6 further being substantially parallel to the front of the tray.
8. The raised ridge of claim 5 further being located at, or forward of, the midpoint of the tray's top surface.
9. The top surface of claim 1 further comprising a raised bump.
10. An apparatus for a pet waste tray, comprising: a tray front to receive pet waste, a tray back end; and a ramp for the tray that holds the tray's front higher than the tray's back end to prevent a first piece of collected pet waste from falling off the tray when a subsequent piece of pet waste is collected.
11. The ramp of claim 10 further comprising a downward angle, the angle being selected from one of a range, wherein the range comprises angles from ten degrees to forty-five degrees, inclusive.

- 12.** The ramp of claim 10, further being of a unitary, one-piece, construction with the pet waste tray.
- 13.** The ramp of claim 10, wherein the ramp is removably attachable to the pet waste tray.
- 14.** The ramp of claim 10, wherein the ramp further comprises an engaging structure to hold the ramp to the tray.
- 15.** The tray of claim 10 further comprising a raised ridge on a top surface of the tray to prevent a first piece of collected pet waste from falling off the tray when the tray is tilted to collect a second piece of pet waste.
- 16.** An apparatus for a pet waste tray, comprising: a raised ridge for a pet waste tray's top surface, wherein the raised ridge is configured to prevent a first piece of collected pet waste located behind the raised ridge from falling off the tray's top surface when a subsequent piece of pet waste is collected onto the tray's top surface.
- 17.** The raised ridge of claim 17 being positioned substantially parallel to a front edge of the tray.
- 18.** The raised ridge of claim 17, wherein ridge is substantially perpendicular to a side wall of the tray.
- 19.** A wall for a pet waste collection tray, comprising: a back wall for a pet waste collection tray, wherein the back wall further comprises a hole that is configured to receive a stream of water from behind the back wall to pass through the hole and clean a surface of the pet waste collection tray.
- 20.** The hole of claim 19, further comprising a size large enough to accommodate a stream of water from a garden hose to pass through it without hitting the back wall.
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